

Exercise sheet 1

Topics in Topology, MTH602

All the spaces below may be assumed to be path connected.

1. What is $\pi_1(X \times Y, (x_0, y_0))$ in terms of $\pi_1(X, x_0)$, and $\pi_1(Y, y_0)$?
2. Show that f_* is injective where $f : S^1 \rightarrow S^1 \times S^1$ defined as $f(t) = (s_0, t)$.
3. Find a covering map $p : S^1 \rightarrow S^1$ which is n -sheeted, which means that the fibre has n points.
4. Show that $\mathbb{R} \times \mathbb{R}$ is a cover of $S^1 \times S^1$ and use that to compute the fundamental group of $S^1 \times S^1$.
5. Show that if $p_1 : X \rightarrow Y$ and $p_2 : Y \rightarrow Z$ are covering maps, then $p_2 \circ p_1$ is a covering map.
6. Show that $p : X \rightarrow Y$ is a covering map, both X and Y are path connected, and Y is simply connected, then p is a homeomorphism.
7. Prove that if $p : X \rightarrow Y$ is a covering then $p_* : \pi_1(X, x_0) \rightarrow \pi_1(Y, y_0)$ is injective.